Airline Route Network
Profitability & Optimization Analysis

Project Overview — Goal, Background & Key Findings

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Portfolio Project — Airline / Aviation Analytics

1. Goal

The goal of this project is to answer a question a real airline network-planning analyst is asked every quarter: **which routes should the airline expand, maintain, watch, or cut?** Instead of looking at one metric in isolation (like total revenue), the project builds a full picture of each route by combining route-level profitability, operational efficiency (load factor), demand trend, and each route's structural importance to the overall network — then turns that into a ranked, actionable recommendation list.

It was also built specifically as a portfolio piece for entry-level data/business analyst roles in the airline and aviation industry, to show both technical range (Python data pipelines, network/graph analysis, dashboarding) and business framing (every output is tied to a decision, not just a chart).

2. How I Came Across This Problem

I've been building out a portfolio of analytics projects (sales forecasting, job market analytics) while job-searching for entry-level data analyst roles, and wanted one project that spoke directly to an industry I'm genuinely interested in breaking into: airlines. Most analytics portfolio projects in this space stop at flight-delay prediction, which is heavily saturated on Kaggle and LinkedIn. Network planning — the question of which routes an airline should fly, add, or drop — is the actual commercial decision airlines make constantly, and it's a much less commonly portfolio-ed problem, which made it a better fit for standing out.

The standard public data source for this kind of analysis is the US DOT/Bureau of Transportation Statistics T-100 Domestic Segment dataset, which every real airline network-planning team uses. This version of the project uses a synthetic dataset generated to match that dataset's structure and realistic route economics (see the Technical Walkthrough PDF and README for details, and for how to swap in the live BTS data).

3. Approach, in Brief

- **Data:** 25-airport network (5 hubs, 20 spokes), 234 routes, 4 quarters of route-level traffic, pricing, and cost data.
- **Metrics engineered:** yield (revenue per passenger-mile), RASM (revenue per available seat mile), operating margin, and a seasonality index — the standard metrics airline network planners use.
- **Network analysis:** modeled the route map as a graph with `networkx` to find each airport's structural importance (centrality) and test network resilience if a hub is disrupted.
- **Scoring model:** combined profitability, load factor, demand growth, and network value into a single Route Health Score, then classified every route into EXPAND / MAINTAIN / MONITOR / CUT relative to the rest of the portfolio.
- **Output:** an interactive dashboard, a ranked action list, and a set of new-route recommendations based on unmet connecting demand.

4. What I Discovered

4.1 The network runs a healthy but uneven portfolio

Metric	Value
Total annual operating profit (network-wide)	$484.8M
Blended operating margin	36.2%
Average load factor	80.3%
Routes flagged EXPAND	47 of 234 (20%)
Routes flagged CUT / RESTRUCTURE	47 of 234 (20%)

The network is profitable overall, but the 20/20 split between EXPAND and CUT candidates shows profitability isn't evenly spread — a fifth of the portfolio is carrying disproportionate value while another fifth is quietly dragging on network efficiency, even though every individual route was profitable in absolute terms.

4.2 Seattle is the most structurally important airport in the network

Graph centrality analysis (eigenvector centrality) ranked Seattle-Tacoma (SEA) as the single most influential airport in the network — ahead of larger legacy hubs like Atlanta and Chicago O'Hare. This wasn't obvious from raw traffic volume alone; it only showed up once the network was modeled as a graph, which is exactly the kind of insight that route-level revenue tables miss.

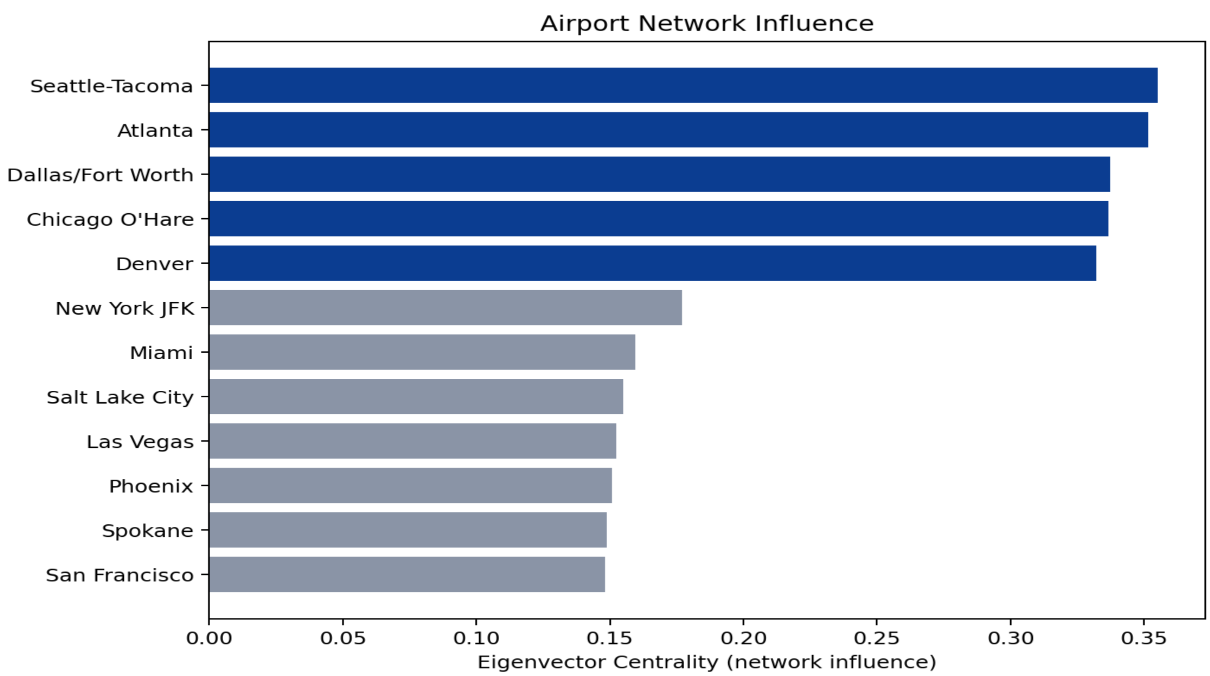

Figure 1 — Airport network influence (eigenvector centrality)

4.3 Profitable routes and healthy routes aren't the same thing

The highest-profit routes (e.g. Seattle–JFK at $7.1M/year) aren't automatically the healthiest. A handful of routes with strong absolute profit (Denver–Kansas City, LA–Chicago) still landed in the bottom margin quintile, because they're being carried by volume rather than efficiency — a distinction that matters for an airline deciding where to invest in additional capacity versus where to hold steady.

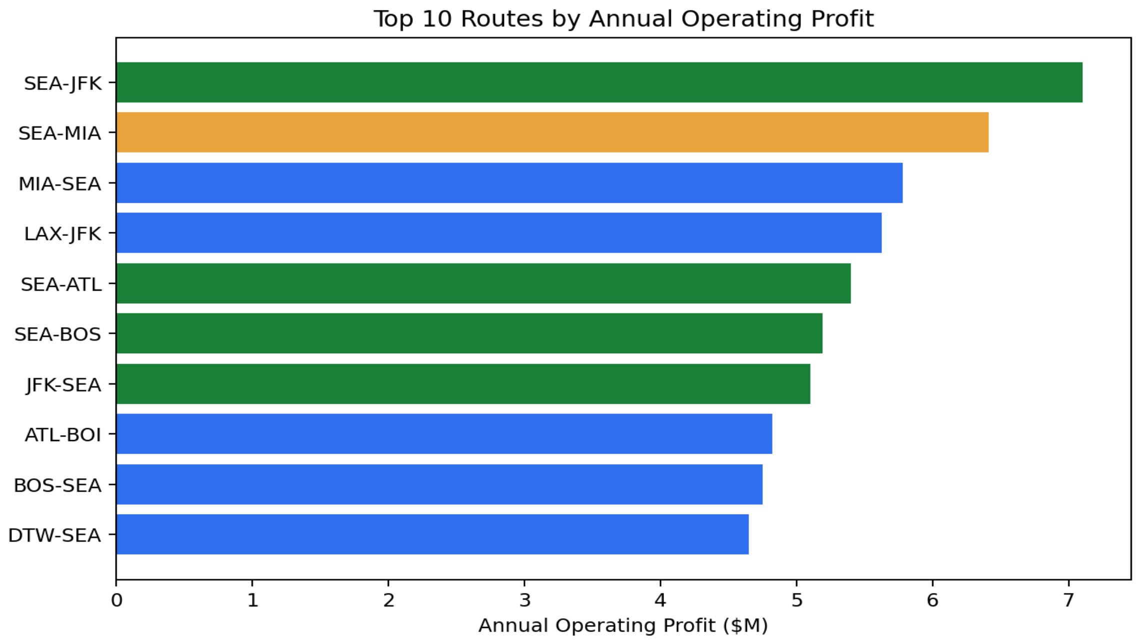

Figure 2 — Top 10 routes by annual operating profit, colored by recommendation

4.4 Demand is seasonal, and it's not uniform across the network

Network-wide passenger volume peaks in Q3 (summer) and dips in Q1, as expected for a US domestic network — but leisure-heavy routes into Las Vegas, Miami, and Phoenix actually peak in Q1 instead, which matters for how the airline should staff and price capacity differently by route type rather than applying one seasonal curve network-wide.

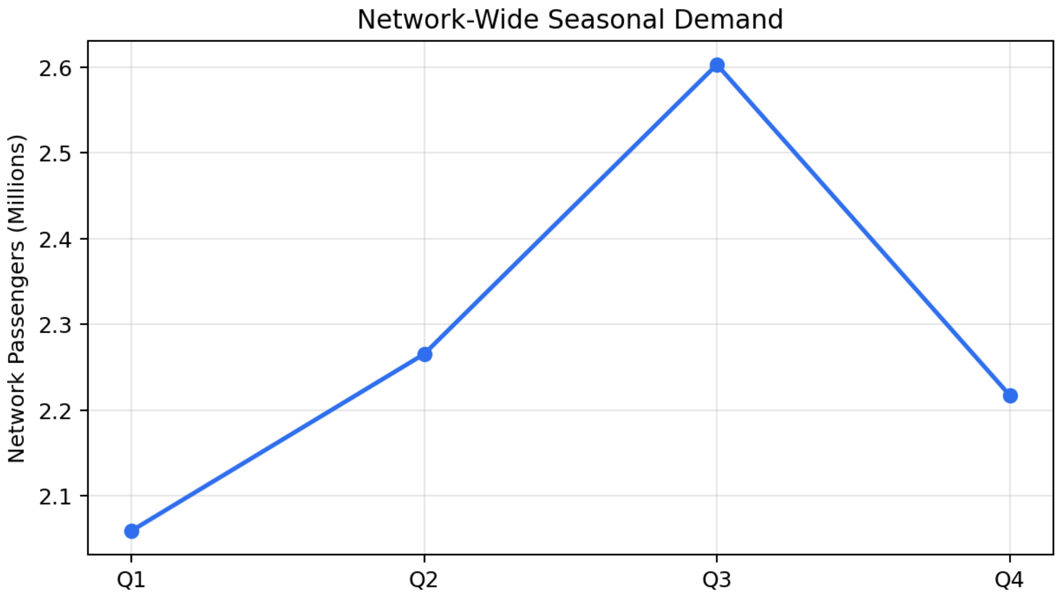

Figure 3 — Network-wide seasonal demand curve

4.5 The network has hub redundancy — but O'Hare is the most exposed

Simulating the removal of each hub showed the network retains 96% connectivity regardless of which hub is disrupted, since the other four hubs still connect to every spoke — a sign of a genuinely resilient hub-and-spoke design. Chicago O'Hare came out as the (marginally) most exposed hub, which is a useful flag for contingency and schedule-resilience planning.

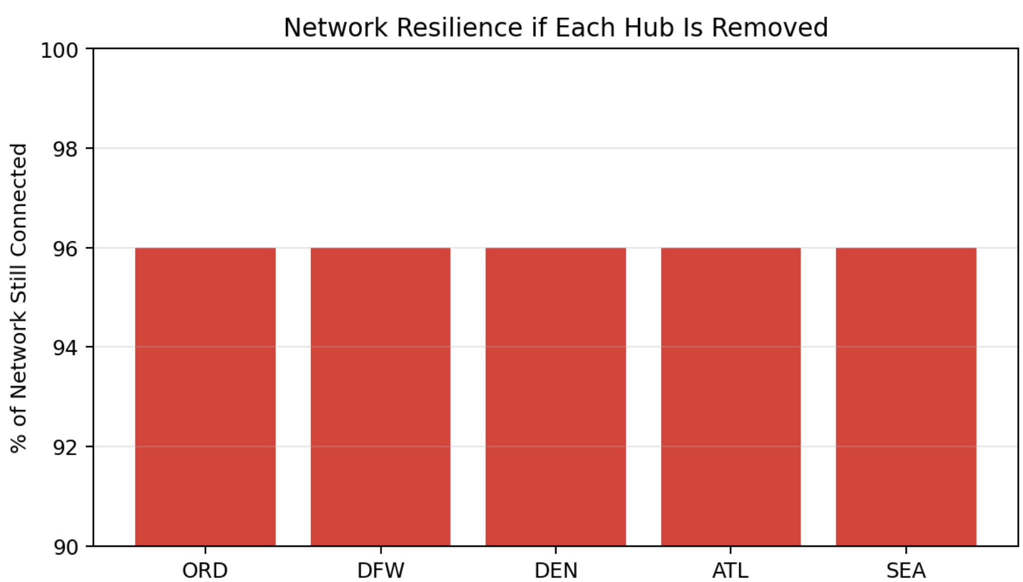

Figure 4 — % of network still connected if each hub is removed

4.6 There's a clear, data-backed case for at least one new route

By looking for spoke-to-spoke city pairs with no existing direct route but high estimated connecting demand through shared hubs, the network graph flagged **Los Angeles – Phoenix** as the strongest new-route candidate (~76,000 estimated annual connecting passengers today with no nonstop option), followed closely by LA–Miami and Miami–Boston.

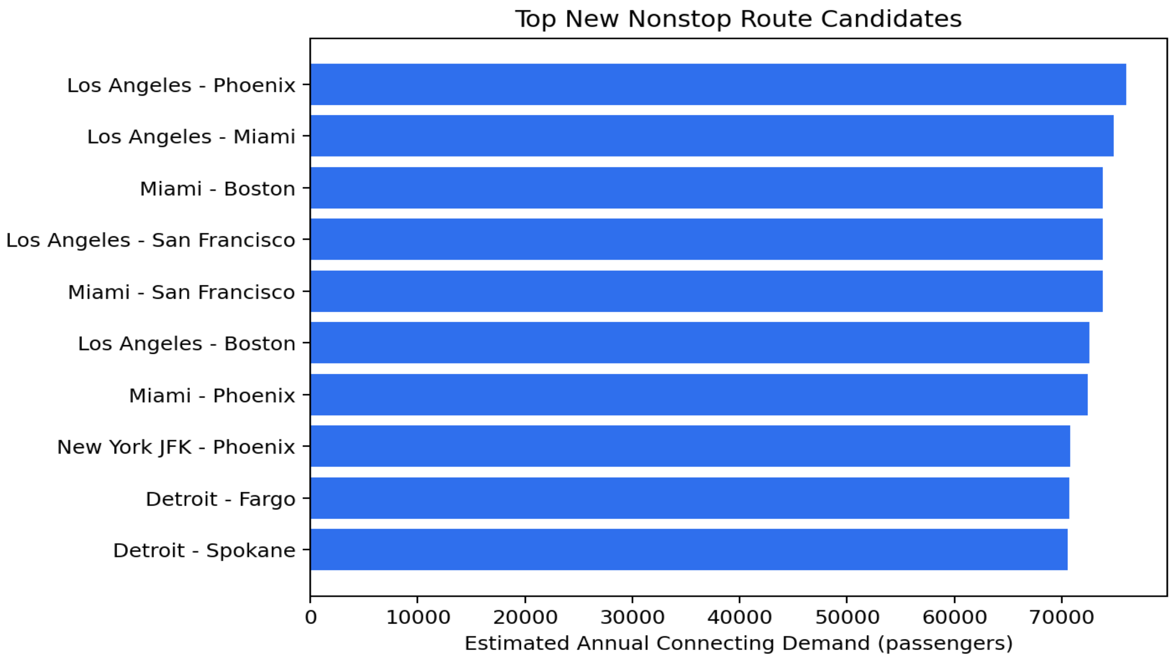

Figure 5 — Top new nonstop route candidates by estimated connecting demand

5. Why This Matters (Business Takeaway)

• A route can be profitable and still be unhealthy — margin and efficiency separate the routes worth investing in from the ones just riding volume.

• Network position (centrality) is invisible in a standard revenue report but changes which routes actually matter strategically — SEA outranking ATL and ORD is the clearest example.

• Seasonality isn't one curve for the whole network — leisure routes and business routes move in opposite directions, and pricing/staffing decisions should reflect that.

• The network already has strong resilience to hub disruption, which is worth confirming rather than assuming.

• Graph-based demand estimation (not just historical route revenue) surfaces new-route opportunities that a simple leaderboard of existing routes would never show, since the opportunity is in a route that doesn't exist yet.

Full technical implementation details are in the companion document, *Technical_Walkthrough.pdf*. All code, data, and the interactive dashboard are included in this project folder.